
Project Title:

Smart Banking Fraud Detection System

Objective:

The primary goal of this project is to design a secure banking system that can:

- Allow users to **register and login securely**.
 - Verify user identity through **PIN authentication**.
 - Perform transactions with **fraud detection mechanisms** to prevent unauthorized or suspicious activities.
 - Maintain a **transaction history** for accountability.
 - Enable users to **update their home location securely**.
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Modules & Features:

1. User Registration:

- Users can create an account with a unique username, password, and home location.
- A **4-digit PIN** is automatically generated for secure operations.
- User data is stored in a **JSON database** (`User_Data.json`).

2. User Login:

- Existing users can log in using their username and password.
- Access to transactions and account features requires login authentication.

3. PIN Verification:

- Sensitive operations, such as transactions or updating home location, require correct PIN entry.
- Prevents unauthorized access even if login credentials are compromised.

4. Fraud Detection & Transactions:

- Users can perform transactions by specifying the amount and location.
- The system checks for:
 - **Unusually large transactions** (e.g., $\geq 1,000,000$ units)
 - **Transactions from locations different from the registered home location**
- Suspicious transactions are **blocked** and logged with a reason.

- All transactions, approved or blocked, are recorded in the user's history.

5. View Transaction History:

- Users can view a complete record of all transactions with details:
 - Date & time
 - Amount
 - Location
 - Status (Approved/Blocked)
 - Reason (if blocked)

6. Update Home Location:

- Users can securely update their home location.
- Requires **PIN verification** and confirmation of the current home location.

7. Data Storage:

- All user information and transactions are stored in a **JSON file** (`User_Data.json`) for persistence.

Tools & Technologies:

- **Programming Language:** Python
- **Data Storage:** JSON
- **Modules Used:**
 - `os` for file handling
 - `json` for storing and retrieving user data
 - `random` for PIN generation
 - `time` for logging transaction timestamps

Conclusion:

The **Smart Banking Fraud Detection System** ensures secure and reliable banking operations by combining user authentication, PIN verification, and fraud detection. It provides transparency through a detailed transaction history while preventing unauthorized access and suspicious activities. This project lays the foundation for a **robust, real-world banking security system** that can be further enhanced with features like multi-factor authentication, alerts for unusual activity, and integration with actual banking platforms.